The Area of Parabolic Segment

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1 Problem

A parabolic segment is a region bounded by a parabola and one of its chords. Let CD be such a chord and let AB be a tangent that is parallel to CD. Connecting AD and BC which is perpendicular to the x-axis to form a parallelogram $ABCD$ shown in Figure 1.

We investigate two questions:

1. Where does the parabola touch the side AB?
2. How is the area of the parabolic segment related to the area of the parallelogram $ABCD$?

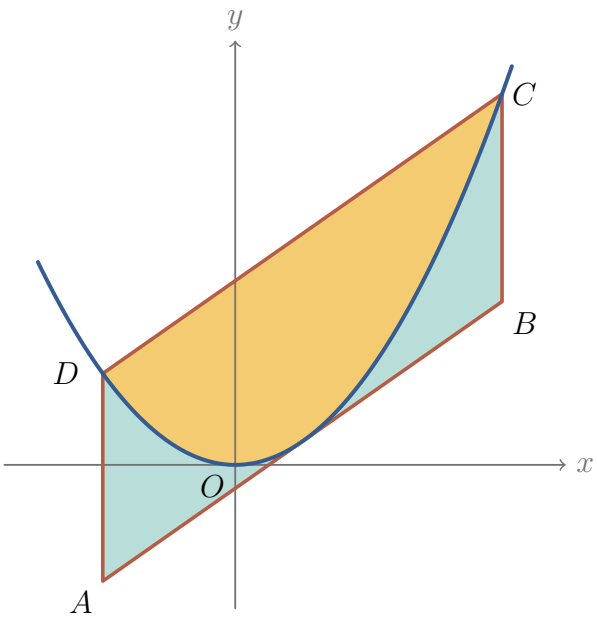


Figure 1: The parabolic segment bounded by the parabola and the chord CD, inscribed in the parallelogram $ABCD$.

2 Solution

Denote the equation of the parabola as $y = ax^2$. We could write the coordinates at D and C as:

$$D : (x_1, ax_1^2) \; and \; C : (x_2, ax_2^2)$$

Without loss of generality, we assume $a > 0$ and $x_1 < x_2$.

First we prove E is the midpoint of AB, $|AE| = |EB|$. Shown in Figure 2.

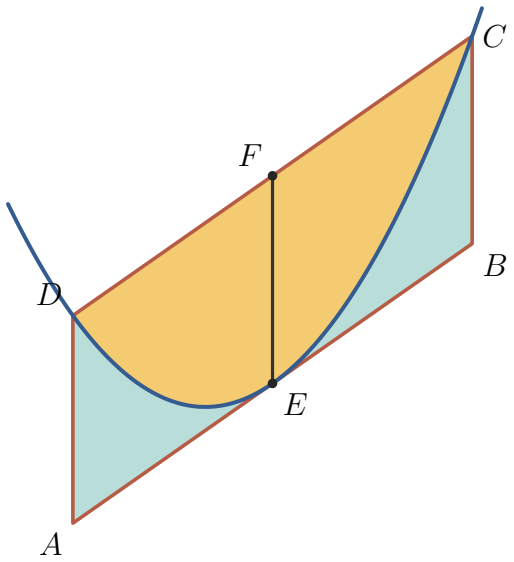


Figure 2: E is the point of tangency between the parabola and the tangent line, and F lies on CD such that EF is perpendicular to the x-axis.

We consider the gradient of CD:

$$m_{CD} = \frac{ax_2^2 - ax_1^2}{x_2 - x_1} = a(x_1 + x_2)$$

Denote the x-coordinate of E is x_E, as AB is a tangent of the parabola, the gradient of AB:

$$m_{AB} = \left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=x_E} = 2ax_E$$

As $AB \parallel CD$,

$$m_{AB} = m_{CD} \implies x_E = \frac{x_1 + x_2}{2}$$

AD and BC is perpendicular to the x-axis, the x-coordinates of A and B are x_1, x_2 respectively. So E is the midpoint of AB.

Then we investigate the relationship between area of the parabolic segment and the parallelogram $ABCD$.

As E is also on the parabola, the coordinates of E, (x_E, y_E), are:

$$\begin{cases} x_E = \frac{x_1+x_2}{2} \\ y_E = ax_E^2 = \frac{a(x_1+x_2)^2}{4} \end{cases}$$

Let us consider the coordinates of F, (x_F, y_F). As E is the midpoint of AB and $EF \parallel AD$, F is also the midpoint of CD due to the property of parallelogram. So the coordinates of F can be expressed using the midpoint formula:

$$\begin{cases} x_F = \frac{x_1+x_2}{2} \\ y_F = \frac{a(x_1^2+x_2^2)}{2} \end{cases}$$

Then we can know the length of line EF:

$$|EF| = |y_F - y_E| = \frac{a(x_1^2 + x_2^2)}{2} - \frac{a(x_1 + x_2)^2}{4} = \frac{a(x_2 - x_1)^2}{4}$$

As $EF \parallel AD$, $AEFD$ is also a parallelogram, $|AD| = |EF|$. We can express the area A_p of parallelogram $ABCD$:

$$A_p = |AD||x_C - x_D| = |EF||x_2 - x_1| = \frac{a(x_2 - x_1)^3}{4}$$

Next, let us find the area of parabolic segment by calculus. Let G be the where AD meet x-axis and let H be the point where the extended CB meet x-axis shown in Figure 3.

The area A_i, bounded by the parabola, DG, CH and x-axis(purple part in Figure 3) equals to:

$$A_i = \int_{x_1}^{x_2} ax^2 dx = \left.\frac{ax^3}{3}\right|_{x=x_1}^{x=x_2} = \frac{a(x_2^3 - x_1^3)}{3}$$

The area, A_t, of trapezium $CDGH$ can also be expressed:

$$A_t = \frac{1}{2}(|DG| + |CH|)|GH| = \frac{a(x_1^2 + x_2^2)}{2}(x_2 - x_1)$$

and the area A_s of parabolic segment (yellow part in Figure 3) equals to:

$$A_s = A_t - A_i = \frac{a(x_1^2 + x_2^2)}{2}(x_2 - x_1) - \frac{a(x_2^3 - x_1^3)}{3} = \frac{a(x_2 - x_1)^3}{6}$$

Finally, the relationship between the area of the parabolic segment(A_s), and the area of parallelogram $ABCD(A_p)$ is:

$$\frac{A_s}{A_p} = \frac{a(x_2 - x_1)^3}{6} \div \frac{a(x_2 - x_1)^3}{4} = \frac{2}{3}$$

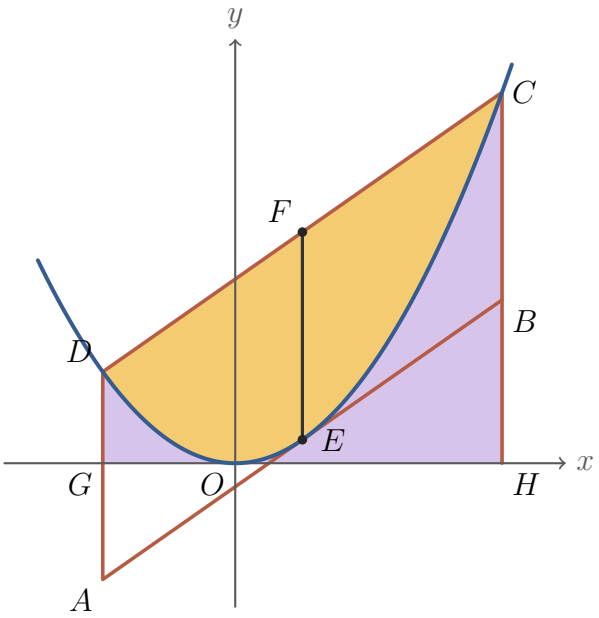


Figure 3: The purple region shows the area under the parabola between G and H on the x-axis.

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Declaration of AI use

The TikZ code used to produce the figures in this document was generated using OpenAI Codex (GPT-6.1 Sol). The code was reviewed by the author.

Source code

The LaTeX source code for this document is available at: `https://github.com/Yijia-phys/mathematical-investigation/tree/main/mini-project-parabolic-segment/`